

# Remix1 + MariaDB on Ubuntu Server

## Docker Compose deployment guide

Prepared for <https://github.com/dyu2x/remix1>

### Read this first

This package is a deployment scaffold. The repository's exact package scripts, database driver, and route structure could not be verified while preparing these files. The Dockerfile assumes npm scripts named `build` and `start` and a server listening on port 3000. The `/connect/admin` path must exist in the Remix application; Compose does not create it.

A browser frontend should never connect directly to MariaDB. The Remix server must connect to the database and serve data/actions through authenticated server routes or APIs.

## 1. Requirements

Use Ubuntu Server 22.04 or 24.04 (64-bit), a sudo-capable account, and internet access. For production, use a domain and HTTPS reverse proxy.

## 2. Install Docker Engine and the Compose plugin

```
sudo apt update
sudo apt install -y ca-certificates curl git
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list >/dev/null
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
sudo systemctl enable --now docker
sudo docker run --rm hello-world
docker compose version
```

If Docker commands require elevated privileges, prefix them with `sudo`. Adding a user to the `docker` group grants root-equivalent control; log out and back in after changing group membership.

## 3. Configure the deployment

Copy the ZIP contents to the Ubuntu server, enter the extracted folder, and create your private environment file:

```
cp .env.example .env
nano .env
```

Set different, strong values for `MARIADB_PASSWORD` and `MARIADB_ROOT_PASSWORD`. Never commit `.env` to Git or share it publicly.

## 4. Build and start the application

```
chmod +x deploy.sh
./deploy.sh
```

The script clones the GitHub repository into `app/`, writes `app/Dockerfile.mesina`, then builds and starts the Remix app and MariaDB containers.

After startup, visit `http://SERVER_IP:3000/`. The admin URL is `http://SERVER_IP:3000/connect/admin` only if the repository implements that route.

## 5. Database configuration

From inside the app container, the database hostname is mariadb, port is 3306, and the database/user/password come from .env. Compose injects DATABASE\_URL and DB\_HOST/DB\_PORT/DB\_NAME/DB\_USER/DB\_PASSWORD. The application code must actually read one of these variables and have a compatible database driver/ORM.

The MariaDB port is intentionally not published on the host. Do not expose port 3306 to the public internet.

## 6. Open the firewall for a basic test

```
sudo ufw allow OpenSSH
sudo ufw allow 3000/tcp
sudo ufw status
```

For production, use HTTPS via Caddy or Nginx and expose only ports 80 and 443 publicly. The direct port-3000 setup is for initial testing.

## 7. Useful management commands

```
docker compose ps
docker compose logs -f app
docker compose logs -f mariadb
docker compose restart app
docker compose down
```

The named database volume persists after docker compose down. Do not run docker compose down -v unless you intend to delete the database volume and all its data.

## 8. Backup and restore

```
mkdir -p backups
set -a
. ./env
set +a
docker compose exec -T mariadb mariadb-dump -u"$MARIADB_USER" -p"$MARIADB_PASSWORD" "$MARIADB_DATABASE" > "backups/backup.sql"
```

To restore, replace backup.sql with your backup filename:

```
set -a
. ./env
set +a
cat backups/backup.sql | docker compose exec -T mariadb mariadb -u"$MARIADB_USER" -p"$MARIADB_PASSWORD" "$MARIADB_DATABASE" < -
```

## 9. Troubleshooting

Build script errors: inspect app/package.json and adjust the Dockerfile to match the real scripts/package manager. Database connection errors: use host mariadb, not localhost, from the app container. A 404 at /connect/admin means the route is missing or named differently in the app. Port conflict: change APP\_PORT in .env. Changing passwords in .env does not automatically update an already initialized MariaDB volume.

## 10. Before production

Confirm app routes and database migrations; implement authentication and authorization for the admin area; configure HTTPS; use unique secrets; keep software updated; and regularly test database backups and restores.